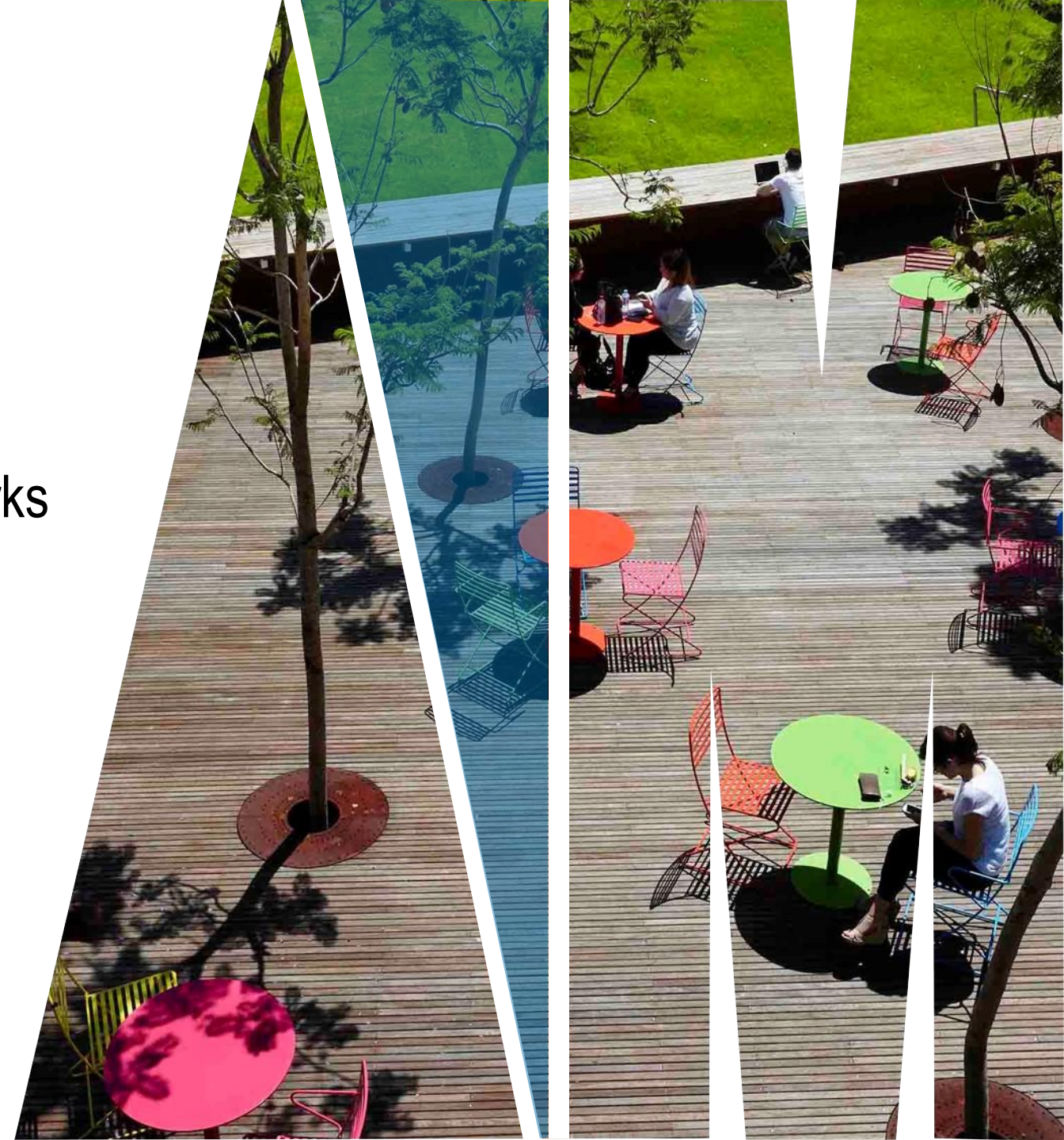
MONASH University

FIT1047

Introduction to computer systems, networks and security

Workshop 2: Boolean Algebra and Digital Circuits



Learning Outcomes

At the end of this workshop, you will:

- be able to create simple digital circuits
- be able to work with Boolean algebra
- be able to understand relation between Boolean logic and digital circuits
- be able to use Logisim to simulate basic circuits
- be able to use Karnaugh-maps to minimise Boolean functions

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Digital Circuits

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Activity 1: 15 mins

Hardware circuits demo

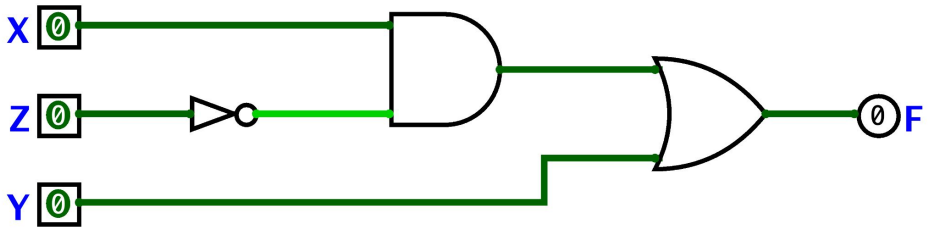
Activity 2: 20 mins

- Work in a small group

  Task 2.1: Discuss some advantages and disadvantages of **simulating a circuit** (as opposed to building it in hardware).

- Task 2.2: Simulate the following circuit in Logisim

$$F(X, Y, Z) = X\overline{Z} + Y$$



- Task 2.3: Try to build an XOR gate only using NAND or NOR gates in Logisim (Don't know how? You can ask Google or ChatGPT first!)

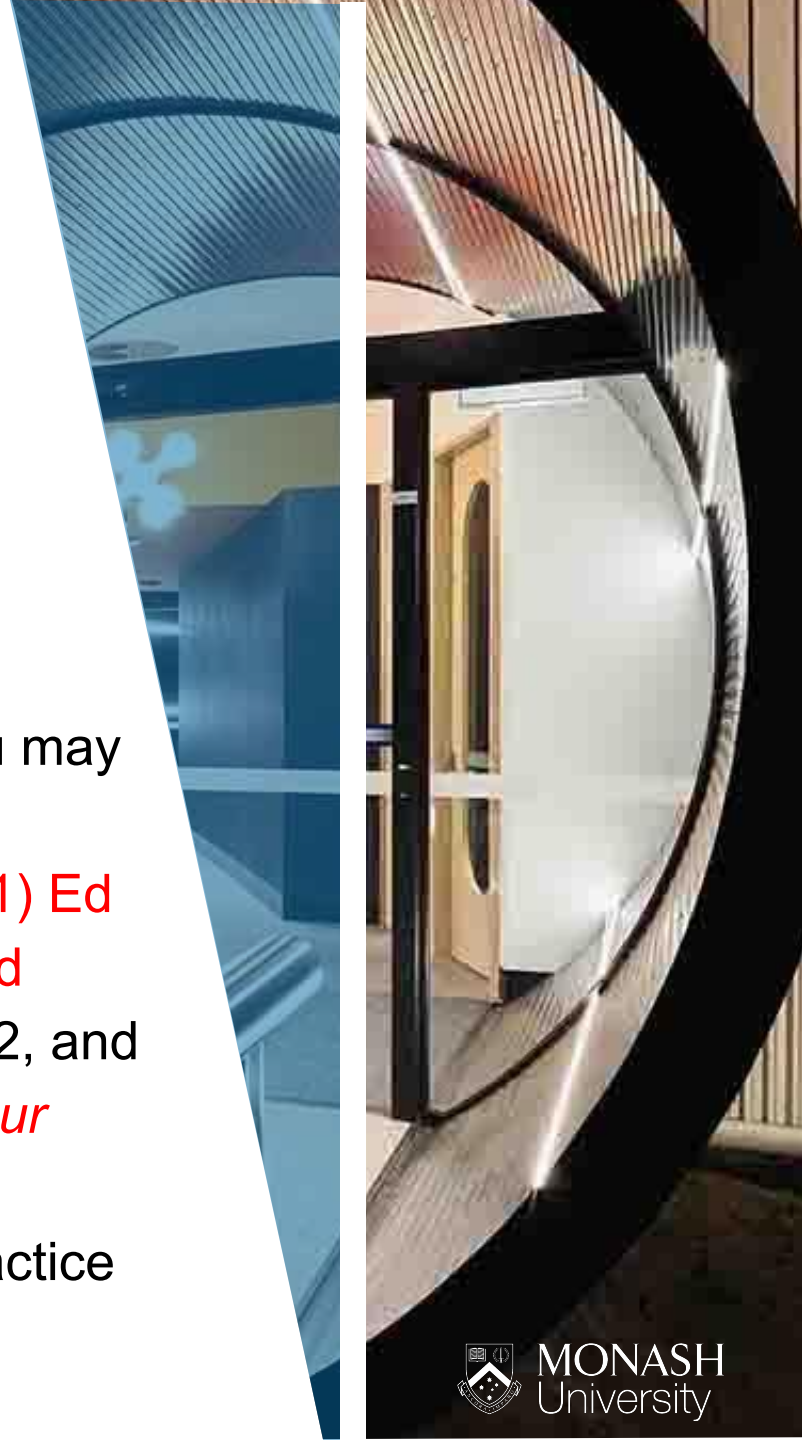
Assignment 1

Assignment 1 requires students to reflect on different ways to represent numbers, and to convert a Boolean truth table into Boolean formulas and optimised circuits.

- Number Systems
- Boolean Algebra

It contains the following parts:

- Moodle submission in Week 3 (specification available now!)
  a. (Part 1) Weekly reflection (at least 200 words per week) – you may copy/paste into a single PDF file for Moodle submission.
     For each week, you must include all three learning activities: 1) Ed lesson (can be the video or pdf notes) 2) Workshop 3) Applied
  b. (Part 2) (1) A PDF file to the answers of the questions in Part 2, and (2) a separate .zip archive with the Logisim files *along with your individualised assignment specification*
- (Part 3) *In-class in-person test in Week 4 Applied class*, with a practice test on Moodle in Week 3.



Assignment 1

Extension and Special Consideration

If you cannot finish on time (and don't want to incur a late penalty):

You can apply for:

1st time: no need to give any reason, and no supporting document is needed
2nd time onwards: need to give reason, but no supporting document is needed

(1) Short Extension (2 days); or
(2) Extension through Special Consideration

Need to provide supporting document.

https://www.monash.edu/students/admin/assessments/extensions-special-consideration

For Part 1 and 2 (Moodle submissions), you can apply for (1) or (2).

For Part 3 (in-class quiz), you **MUST** apply for (2) Extension through Special Consideration if you miss the quiz or cannot attend on that day.

Make sure to apply **as soon as possible, but no later than 11.55pm on the day your assessment is due**.

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Break: 10 mins

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Recap: Karnaugh Maps

Karnaugh Maps (K-Maps)

3-Variable K-Map:

XY \ Z	0	1
00	0	0
01	1	1
11	1	1
10	1	0

Note: 11 before 10!

transfer

X	Y	Z	F(X,Y,Z)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

K-map rules

- Groups can’t contain 0

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Correct

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Wrong

K-map rules

- Groups must be rectangular (can’t be e.g. diagonal)

XY \ Z	0	
00	0	0
01	0	1
11	1	1
10	0	0

Correct

XY \ Z	0	
00	0	0
01	0	1
11	1	1
10	0	0

Wrong

K-map rules

- Group size is power of 2

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Correct

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Wrong

K-map rules

- Each group is as large as possible
- Groups may overlap

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Correct

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Wrong

K-map rules

- Each 1 must be in at least one group

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Correct

XY \ Z	0	
00	0	0
01	1	1
11	1	1
10	1	0

Wrong

K-map rules

- Groups may wrap around

XY \ Z	0	
00	1	1
01	0	0
11	0	0
10	1	1

Correct

XY \ Z	0	
00	1	1
01	0	0
11	0	0
10	1	1

Wrong

3-variable K-Map

XY \ Z	0	1
00	0	0
01	1	1
11	1	1
10	1	0

X and Z are irrelevant (group covers both 0 and 1 for these variables).
Y needs to be 1 for this group.

$$\overline{X}\,Y\,\overline{Z} + \overline{X}\,Y\,Z + X\,Y\,\overline{Z} + X\,Y\,Z = Y$$

Y is irrelevant (group covers both 0 and 1 for Y).
X needs to be 1, Z needs to be 0 for this group.

$$X\,Y\,\overline{Z} + X\,\overline{Y}\,\overline{Z} = X\,\overline{Z}$$

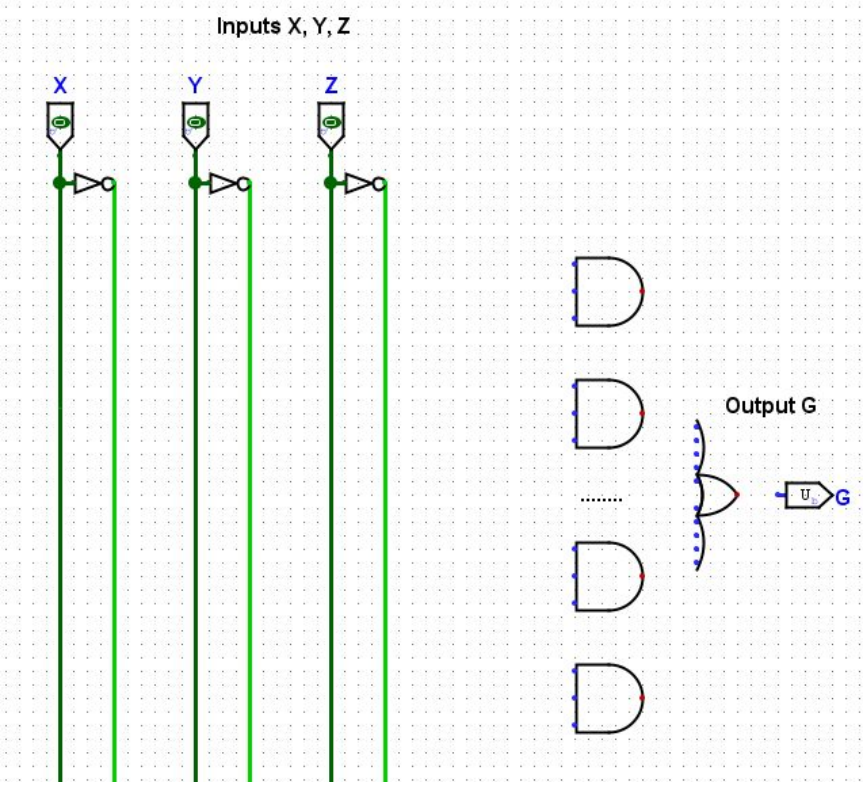
Result: $F(X,Y,Z) = X\overline{Z} + Y$

Boolean Expressions

Activity 3 Tasks: 10 mins

- Work in a small group
- Task 3.1: Simulate the circuit in Logisim

G(X,Y,Z) = $\bar{X}\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}\bar{Z} + XY\bar{Z} + XYZ$



Reflection opportunity!

You may use the template provided in Moodle!

Simplification

Activity 4 Tasks: 20 mins

- Work in a small group
- Task 4.1: Simplify the circuit that you built in Task 3.1

  G(X,Y,Z)= $\bar{X}\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}\bar{Z} + XY\bar{Z} + XYZ$

  using a Karnaugh map (K-map)
- Task 4.2: Use Logisim to build your simplified circuit (in Task 4.1) and verify the result with your previous circuit (built in Task 3.1).

Reflection opportunity!

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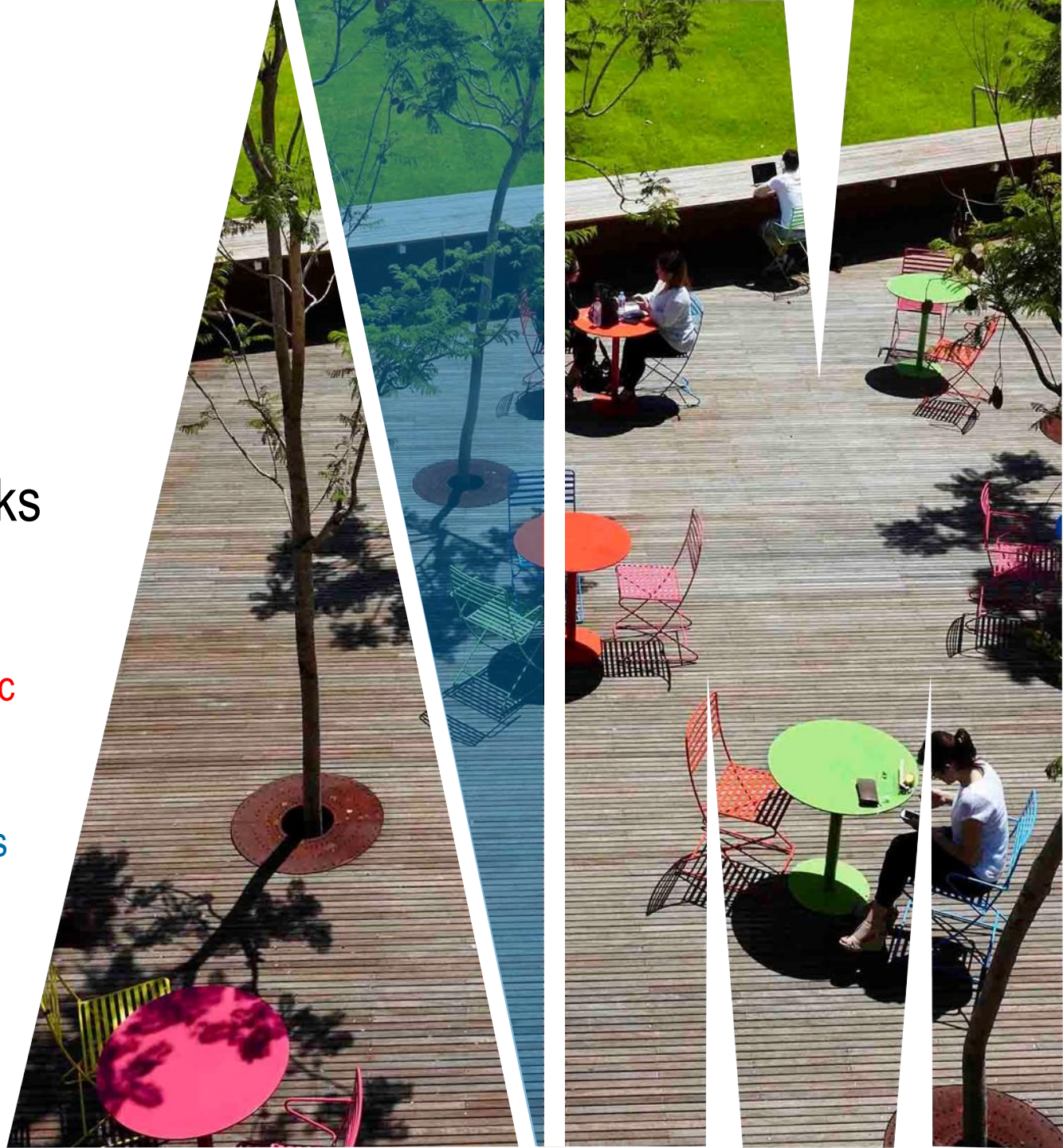
Introduction to computer systems, networks and security

Workshop 2: Boolean algebra, digital circuits, Boolean logic and digital circuits, and Karnaugh-maps

Before next week's workshop: Weekly videos and readings

See you next week!